Nithik Vairamuthu

Architect & Computational Designer

AI tools for buildings, cities and habitats

Computer vision · Generative AI · Interactive systems · Hardware

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Master's student in AI for Architecture and the Built Environment at IAAC Barcelona, building tools for architects, engineers and urban planners. I work end to end: computer vision and machine learning pipelines in Python, the data behind them, and the interfaces and hardware around them.

SELECTED PROJECTS

2026

OASIS — Best Thesis Award

Main tool developer · team of 4, IAAC MaAI · Tutor: Prof. Emanuele Naboni

AI assisted retrofit tool for Barcelona's extreme heat. It scores heat vulnerability for every building in a drawn zone, then traces it down to room level retrofits ranked from climate, building and health data. Built the complete urban scale tool and integrated the team's workflows into one system.

Stack Infrared SDK · UTCI · Ollama / Qwen2.5 · IFC · OpenStreetMap · Gemini API

2026

RetroGrade

Tool developer · team of 4, IAAC Research Studio II

Retrofit feasibility score out of 100 from one façade photo and a short questionnaire, in about 60 seconds. Built the image segmentation step (SAM, SegFormer, CLIP) that extracts window to wall ratio, shading, vegetation and material. A random forest on 40,000 European buildings predicts energy and carbon (MAE 7.79, 6.90).

Stack SAM · SegFormer · CLIP · Random Forest · Open-Meteo · ReportLab

2026

RE:PAIR

Developer, crack detection · team of 5, IAAC MRAC / MaAI

Closed loop robotic repair of cracked timber: detect, repair by robotic 3D printing, verify. Built the custom trained YOLO segmentation and the code that turns masks into centrelines and UR10 targets, plus a Flask bridge that serves it to Grasshopper. Tested in RViz and on a real UR10.

Stack Python · YOLOv8 · ROS 2 · UR10 · RViz · Flask · Grasshopper

2025

Chronoscape

Lead developer · IAAC Research Studio

Turns residents' memories of a lost Pacific Palisades neighbourhood into weighted prompts for AI reconstruction of a home and a shared street. Wrote the full pipeline: narrative parser, emotion analysis, location enhancer and image generation. 13 narratives, 250 images, 82 elements.

Stack Python · custom parser · pypdf · emotion analysis · Gemini (Nano Banana)

2026

Guided Comparative Assembly

Data, UI & workflows · IAAC MRAC + MaAI · Hardware III

Tabletop installation where visitors configure a building and see the carbon, labour, time and cost of masonry, 3D printed and prefab construction projected back onto the table, every figure a sourced range. Worked on data gathering, the nine panel projected UI, the presentation and the workflows connecting the system. Presented at the IAAC final critique, 22 May 2026.

Stack TouchDesigner · Python · OpenCV · MediaPipe · ESP32 + RFID · OSC

2026

Pressure Chamber · exhibited

Coding & installation · Barcelona, World Capital of Architecture 2026

Interactive installation where visitors try to out prompt an image generator and discover its biases first hand; CLIP scores each attempt against a target. Built in three languages with an offline fallback, and run as a kiosk on a Raspberry Pi.

Stack Python · Flask · Gemini API · CLIP ViT-B/32 · React · Three.js · Raspberry Pi

SELECTED PROJECTS, CONTINUED

2026
MVP

Pawlytics

Developer · group project

Pet health monitoring app that compares each pet with its own 30 day baseline, adds breed risk and returns a 0–100 score with Monitor / Vet soon / Urgent guidance. Worked on the FastAPI and PostgreSQL backend, the rules based health engine and the React front end with a 3D avatar.

Stack React · TypeScript · Three.js · FastAPI · SQLAlchemy · PostgreSQL · JWT

RESEARCH & PUBLICATION

Nov 2025

Computational Study of Origami Inspired Geometries in Lunar Habitations

Presented at the ZMECH Conference, VIT Vellore

Dec 2023

LUNA: The Moon Base

Undergraduate thesis on settlements on the lunar surface

EDUCATION

Oct 2025 – Jul 2027

Institute for Advanced Architecture of Catalonia (IAAC)

Master's in AI for Architecture and the Built Environment · Merit scholarship, 2025

Oct 2025 – present

EADA Business School

Business Strategist, Business Administration and Management

Vellore

Vellore Institute of Technology

Bachelor of Architecture (B.Arch) · CGPA 8.80 / 10 · University rank holder, B.Arch

EXPERIENCE

Teaching

Teaching Assistant

Institute for Advanced Architecture of Catalonia (IAAC), Barcelona

Teaching assistant for computational design to first year Master's in AI students.

Dec 2022 – Apr 2023

Architectural Internship

Iinfinity Architects, Chennai

Commercial and residential projects and project documentation workflows.

Tutor

Computational Design Workshop

Mysore School of Architecture

Conducted a 3 day computational design workshop.

Aug 2026
Workshop

AI Workshop

VIT University, Vellore

Workshop

VAP Parametric Workshop

VIT University, Vellore

Parametric modelling, fabrication and furniture design.

SKILLS

AI & vision	Python · PyTorch · OpenCV · SAM · SegFormer · YOLOv8 · CLIP · MediaPipe · Stable Diffusion · regression models
Comp. design	Grasshopper · Rhino · Speckle · parametric design · TouchDesigner
Software	React · TypeScript · FastAPI · Flask · PostgreSQL · Three.js
Hardware	Arduino · ESP32 · RFID · sensor integration · ROS 2 · UR10 · Raspberry Pi
Languages	Tamil (native) · English (fluent)
Interests	AI assisted workflows · extreme and extraterrestrial habitats · interdisciplinary collaboration